Numerical analysis of superconducting phases in the extended Hubbard model with non-local pairing

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Thesis for the Master's degree in Physics

Abstract

[To be continued...]

- Appendix on s-wave SC ghost simulations on HM
- Appendix on triplet pairing in MFT self-consistent scheme
- Set ComputerModern font in all CairoMakie figures

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List of symbols and abbreviations

AF	Antiferromagnetic
BCS	Bardeen-Cooper-Schrieffer (theory)
BZ	Brillouin zone
β	Inverse temperature
$\Delta^{(\gamma)}$	γ-wave component of the gap function
DoF	Degree of freedom
EHM	Extended Hubbard model
EV(s)	Expectation value(s)
FS	Fermi surface
HF	Hartree-Fock
HFP(s)	Hartree-Fock parameter(s)
HM	Hubbard model
L_ℓ	Number of lattice sites in direction ℓ
iHF	Iterative Hartree-Fock
LRT	Linear Response Theory
MBZ	Magnetic Brillouin zone
MFT	Mean-Field Theory
NBZ	Nematic Brillouin zone
NN(s)	Nearest neighbor(s)
RWC(s)	Relevant Wick's contraction(s)
RMP(s)	Renormalized model parameter(s)
SC	Superconductor
SDW	Spin-density wave
SSB	Spontaneous symmetry breaking
T_c	Critical temperature

Introduction

This thesis project is about my favorite ice cream flavor. [To be continued. . .]

Chapter 1

Mean-field theory discussion of the EHM

[Introduction: this is a technical chapter about Hartree Fock applied to EHM.]

1.1 Extended Hubbard model symmetries and Wick's theorem

The general aim of this project is to study the phase diagram of the EHM by comparing ground-state energies of different phases. The phases we consider in next chapters for the EHM are the normal state (N), the antiferromagnetic ordering (AF), given by a non-uniform distribution of charge in each spin sector, and the anisotropic superconducting (SC) phase, described by a uniformly distributed charge allowing for Cooper pairing instabilities.

At the end of this section, in Tab. 1.2 we summarize all symmetries with the relative operations. Throughout the entire thesis we will refer to specific SSB for given groups to establish a specific phase.

1.1.1 Spatial and topological symmetries

Let us start from the trivial topology of the lattice we are working on. The EHM of Eq. (**??**) is evidently translational invariant on both the x and y directions. We refer to these groups as $\mathrm{T}_1^{(x)}$, $\mathrm{T}_1^{(x)}$. The "1" subscript indicates the one-site translations under which the model is symmetric.

Evidently, the following statement holds:

$$\mathrm{T}_n^{(\ell)} \supset \mathrm{T}_{kn}^{(\ell)} \qquad k \in \mathbb{N}$$

A model which is invariant for n-sites translations in the ℓ direction, also is for kn-sites translations. Then we can break translation symmetry in a given direction reducing it to a smaller group. For example, the commensurate AF phase we analyze halves translational invariance,

$$\bigotimes_{\ell=x,y} \mathrm{T}_1^{(\ell)} \overset{\mathrm{SSB}}{\rightarrow} \bigotimes_{\ell=x,y} \mathrm{T}_2^{(\ell)}$$

since it deals with two-site periodic spin-density waves (SDWs); instead the SC phase preserves translational invariance.

Another natural symmetry of the EHM is the four-fold point group C_4 of $\pi/2$ rotations, expressing the square lattice property being mapped onto itself under said rotations. A similar property as for the translations holds for point groups,

$$\mathrm{C}_n \supset \mathrm{C}_{n/k} \qquad k, n/k \in \mathbb{N}$$

which expresses the fact that, if a system is topologically invariant under n-fold rotations and $k \in \mathbb{N}$ exists such that $n/k \in \mathbb{R}$, then the smaller point group is automatically included. As for translations, we can break the point group reducing it to a smaller subgroup, say

$$\mathrm{C}_4 \overset{\mathrm{SSB}}{\rightarrow} \mathrm{C}_2$$

Phase	SSBs
AF	$\mathrm{T}_1^{(\ell)} \to \mathrm{T}_2^{(\ell)}$
	$\mathrm{SU}(2) \to \mathrm{U}^z(1)$
SC	$\mathrm{U}^c(1) \to 0$

Table 1.1 SSB interpretation of the AF and SC phases. In the last line, the 0 on the SSB right-hand side indicates a completely disrupted symmetry.

This is what happens in general with the effective arising of anisotropic corrections to the kinetic hamiltonian due to non-local interactions, as will be discussed in Sec. **??**. For our purposes, we will leave C_4 unbroken both in the AF and SC phases.

Finally, the system also is inversion-symmetric: let I_2 be the inversion group on the two-dimensional square lattice, containing trivially just the inversion operator and the identity. In general, rotations and mirroring are different operations from a group-theoretical point of view. However, when it comes to the square lattice, operatively it holds

$$\mathrm{I}_2 \equiv \mathrm{C}_2$$

To rotate by π is the same as to mirror the lattice. The inversion symmetry is then already included in C_4, and we leave it unspecified.

1.1.2 Spin and charge symmetries

When it comes to Fermi algebra and operators, a large symmetry of the EHM is a global U(2) symmetry [9] given by the transformation

$$\hat{c}_{i\sigma} \to \sum_{\sigma'} \mathcal{U}_{\sigma\sigma'} \hat{c}_{i\sigma'} \quad \text{with} \quad \mathcal{U}_{\sigma\sigma'} \in \mathrm{U}(2) \tag{1.1}$$

The proof is trivial and follows from the fact that $\mathcal{U}^\dagger \mathcal{U} = \mathcal{U}\mathcal{U}^\dagger = \mathbb{I}$ [add proof?]. Note that the the following decomposition holds,

$$\mathrm{U}(2) = \mathrm{SU}(2) \otimes \mathrm{U}^c(1) \tag{1.2}$$

which expresses the separate charge conservation and spin rotational symmetries. In detail:

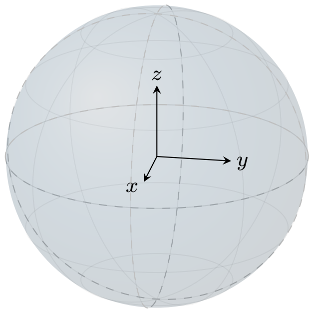
- $\mathrm{U}^c(1)$ charge conservation, coming from Eq. (1.2) decomposition. By breaking this symmetry, we allow for the total charge to fluctuate in the ground state. These fluctuations are only allowed in the SC phase in the present text;

- SU(2) spin rotation symmetry, also , coming from Eq. (1.2) decomposition. Note that:

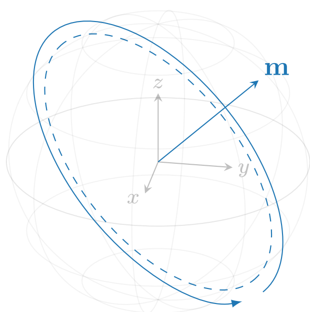
$$\mathrm{SU}(2) \supset \mathrm{U}^z(1)$$

  The z subscript we included to distinguish from the charge group. Hence SU(2) symmetry can be broken, selecting a particular vectorial direction for the order parameter (we call z), and reduced to a smaller $\mathrm{U}^z(1)$ symmetry which is essentially expressed by the conservation of the magnitude of the order parameter.

The last point is important: for the AF phase, for example, there is no need for the magnetization vector to be directed along the physical z-axis, which is the one perpendicular to the lattice plane. SU(2) symmetry breaks when (staggered) magnetization is established along a particular direction, but spin rotations around said direction still are ground state symmetries. This is shown schematically in Fig. 1.1.



a | SU(2) rotations domain before SSB.



b | U(1) rotations domain after SSB.

Figure 1.1 | Graphic representation of the concept of SU(2) $\to$ U^z(1) SSB. The solid color identifies, in both cases, the rotations domain for the current symmetry. When SSB takes place in a magnetic phase, the magnetization order parameter acquires a precise direction which was previously invariant over the full spherical domain. After SSB, the hamiltonian is rotationally invariant only for rotations *around* the selected magnetization vector.

1.1.3 Particle-hole symmetry at half-filling

The half-filled EHM is of particular interest due to its particle-hole symmetry (PHS), which is immediately distrupted at infinitesimal doping. Let us prove that. Consider the generic unitary operation

$$\mathrm{P} \quad : \quad \hat{c}_{i\sigma} \quad \to \quad e^{i\eta_{i\sigma}}\hat{h}^\dagger_{i\sigma}$$

being in general η_i a phase dependent on the fermionic quantum state $(i\sigma)$. Inserting the above transformation in the EHM, we get

$$\mathrm{P}\{\hat{H}\} = -t\sum_{\langle ij\rangle}\sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma})}\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma} + U\sum_i \hat{h}_{i\uparrow}\hat{h}_{i\downarrow}\hat{h}^\dagger_{i\downarrow}\hat{h}^\dagger_{i\uparrow} - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'}\hat{h}^\dagger_{i\sigma}$$

For the two interactions $\hat{H}_U$, $\hat{H}_V$ the phase factors cancelled out. We aim to find the conditions under which the above hamiltonian is mapped onto the EHM of (**??**),

$$\mathrm{P}\{\hat{H}\} \stackrel{?}{=} -t\sum_{\langle ij\rangle}\sum_{\sigma}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma} + U\sum_i \hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma}$$

Before starting the derivation, which is an extension of the argument of Arovas et al. [9] about the conventional HM, let us discuss some properties of the hole operators. Due to the presence of the phase factors in the PHS operative definition, the Fermi anticommutation rules hold regardlessly of the phase factor:

$$\begin{aligned}\{\hat{h}_{i\sigma},\hat{h}^\dagger_{j\sigma'}\} &= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})}\{\hat{c}^\dagger_{i\sigma},\hat{c}_{j\sigma'}\} \\ &= e^{i(\eta_{i\sigma}-\eta_{j\sigma'})}\delta_{ij}\delta_{\sigma\sigma'} \\ &= \delta_{ij}\delta_{\sigma\sigma'}\end{aligned}$$

and evidently:

$$\hat{h}_{i\sigma}\hat{h}^\dagger_{i\sigma} = \hat{n}_{i\sigma}$$

being $\hat{n}_{i\sigma}$ the fermionic number operator on site i with spin σ. It follows, for quartic interactions

$$\begin{aligned}
\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'}\hat{h}^\dagger_{i\sigma} &= -\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'} \\
&= \hat{h}_{i\sigma}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= -\hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'}\delta_{ij}\delta_{\sigma\sigma'} \\
&= -\hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{i\sigma}\hat{h}_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{i\sigma} + \hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'} - \left(\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'} \\
&= \hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} + (\hat{n}_{i\sigma} + \hat{n}_{j\sigma'}) - 1 - \left(\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma'} - \hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma'}\right)\delta_{ij}\delta_{\sigma\sigma'}
\end{aligned}$$

The first term of the last passage is mirrored with respect to the starting term. Now, let us deal with the kinetic, local and non-local interactions separately.

Kinetic term. Consider the kinetic term,

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma})}\hat{h}_{i\sigma}\hat{h}^\dagger_{j\sigma}$$

It suffices to anticommute the two fermionic operators, and exchange site labels $i \leftrightarrow j$, to get

$$\mathrm{P}\{\hat{H}_t\} = -t\sum_{\langle ij\rangle}\sum_{\sigma} e^{i(\eta_{i\sigma}+\eta_{j\sigma}+\pi)}\hat{h}^\dagger_{i\sigma}\hat{h}_{j\sigma}$$

A necessary condition on the unitary operation P is the following,

$$\eta_{i\sigma} + \eta_{j\sigma} = (2k-1)\pi \quad \text{with} \quad k \in \mathbb{Z}$$

valid for NN sites.

Local repulsion. Substitute in $\hat{H}_U$ the above derivation,

$$\begin{aligned}
\mathrm{P}\{\hat{H}_U\} &= U\sum_i \hat{h}_{i\uparrow}\hat{h}_{i\downarrow}\hat{h}^\dagger_{i\downarrow}\hat{h}^\dagger_{i\uparrow} \\
&= U\sum_i \hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U\sum_i(\hat{n}_{i\uparrow}+\hat{n}_{i\downarrow}) - U\sum_i(1) \\
&= U\sum_i \hat{h}^\dagger_{i\uparrow}\hat{h}^\dagger_{i\downarrow}\hat{h}_{i\downarrow}\hat{h}_{i\uparrow} + U(\hat{N} - L_xL_y)
\end{aligned}$$

Thus, being for the half-filled state

$$\langle\hat{N}\rangle = L_xL_y$$

preserves PHS in the local sector.

Non-local attraction. Proceed identically for $\hat{H}_V$,

$$\begin{aligned}
\mathrm{P}\{\hat{H}_V\} &= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'} \hat{h}_{i\sigma}\hat{h}_{j\sigma'}\hat{h}^\dagger_{j\sigma'}\hat{h}^\dagger_{i\sigma} \\
&= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'} \hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} - V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}(\hat{n}_{i\sigma}+\hat{n}_{j\sigma'}) + V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'}(1) \\
&= -V\sum_{\langle ij\rangle}\sum_{\sigma\sigma'} \hat{h}^\dagger_{i\sigma}\hat{h}^\dagger_{j\sigma'}\hat{h}_{j\sigma'}\hat{h}_{i\sigma} - 2zV(\hat{N} - L_xL_y)
\end{aligned}$$

with $z = 4$ the lattice coordination number. In the last passage, we used that on the square lattice

$$\begin{aligned}
\sum_{\langle ij\rangle}(1) &= \sum_{i\in\mathcal{S}/2}\sum_{j\in\mathrm{NN}(i)}(1) \\
&= \frac{L_xL_y}{2}\times z
\end{aligned} \tag{1.3}$$

Symmetry	Operations
$\mathrm{T}_1^{(x)} \otimes \mathrm{T}_1^{(y)}$	One-site discrete translations separately in both directions
C_4	Four-fold point group (real $\pi/2$ angle rotations)
$\mathrm{U}^c(1)$	Charge-conserving global phase shifts
SU(2)	Three-dimensional rotations in spin space
PHS	Particle-hole exchange (only at half filling)
$\mathrm{U}^z(1)$	Reduced two-dimensional rotations in spin space

Table 1.2 Model symmetries and relative group operations. Note that $\mathrm{U}^c(1)$ represents the charge conserving group, while $\mathrm{U}^z(1)$ represents the subgroup obtained by breaking the SU(2) symmetry with a vector order parameter and preserving its amplitude. Since the latter is "derived" from SU(2), we separate it from the four main symmetries of the model.

which will turn out quite often. As before, an half-filled phase preserves PHS in the non-local sector. In last passage we used the fact that summing over NNs means overcounting each site $z/2$ times, and spin degeneracy.

No other condition on phases is derived, thus we conclude the half-filled EHM preserves PHS if the two sublattices $\mathcal{S}_a$, $\mathcal{S}_b$ the square lattice can be divided into, being a (trivial) bipartite lattice, differ by a π phase difference under PHS transform. In other words, if P is defined by the unitary operation

$$\mathrm{P} \quad : \quad \begin{cases} \hat{c}_{i\sigma} \to \hat{h}_{i\sigma}^\dagger & \text{for} \quad i \in \mathcal{S}_a \\ \hat{c}_{i\sigma} \to -\hat{h}_{i\sigma}^\dagger & \text{for} \quad i \in \mathcal{S}_b \end{cases}$$

and the system is half-filled, PHS is respected. Any other phase choice, as long as it alternates by π on the two sublattices, is good enough. To summarize, we report in Tab. 1.2 all identified symmetries.

1.2 MFT on Hubbard lattices

This section serves as groundwork for everything that follows. We here define the MFT scheme as well as the iterative HF computational scheme we use to estimate parametrically the relevant physical quantities of the various phases. We start by recalling the general features of MFT and then move to the specificity of the EHM, apply Wick's theorem and finally move to algorithic aspects.

1.2.1 MFT general procedure, self-consistency and Wick's theorem

The Mean Field Theory approach to any fermionic quartic operator is given by the following procedure: given the hamiltonian made of the quadratic operators $\hat{A}$ and $\hat{B}$

$$\hat{H} \equiv \hat{A}\hat{B}$$

we define the fluctuations

$$\delta\hat{A} \equiv \hat{A} - \langle\hat{A}\rangle \qquad \delta\hat{B} \equiv \hat{B} - \langle\hat{B}\rangle$$

thus the following chain holds:

$$\begin{aligned} \hat{A}\hat{B} &= \langle\hat{A}\rangle\langle\hat{B}\rangle + \delta\hat{A}\langle\hat{B}\rangle + \langle\hat{A}\rangle\delta\hat{B} + \delta\hat{A}\delta\hat{B} \\ &= -\langle\hat{A}\rangle\langle\hat{B}\rangle + \hat{A}\langle\hat{B}\rangle + \langle\hat{A}\rangle\hat{B} + \underbrace{\delta\hat{A}\delta\hat{B}}_{\mathcal{O}(\delta)^2} \end{aligned}$$

which implies

$$\langle\hat{H}\rangle = \langle\hat{A}\rangle\langle\hat{B}\rangle + \langle\delta\hat{A}\delta\hat{B}\rangle$$

Thus, if $|\langle\hat{A}\rangle\langle\hat{B}\rangle| \gg |\langle\delta\hat{A}\delta\hat{B}\rangle|$, in a statistical sense we can neglect the $\mathcal{O}(\delta)^2$ part of the hamiltonian and approximate

$$\hat{H} \simeq \hat{A}\langle\hat{B}\rangle + \langle\hat{A}\rangle\hat{B} - \langle\hat{A}\rangle\langle\hat{B}\rangle \qquad \Longrightarrow \qquad \langle\hat{H}\rangle \simeq \langle\hat{A}\rangle\langle\hat{B}\rangle \tag{1.4}$$

Term	Normal phase	Antiferromagnet	Superconductor
Hartree			
Fock	Prohibited	Prohibited	Prohibited
Cooper	Prohibited	Prohibited	

Table 1.3 Summary of Wick's terms selection rules and relevant contributions to the effective hamiltonian for each phase limitedly to $\hat{H}_U$.

Self consistency of MFT procedure is thus embodied by the condition

$$\left|\frac{\langle\delta\hat{A}\delta\hat{B}\rangle}{\langle\hat{A}\rangle\langle\hat{B}\rangle}\right| \ll 1 \qquad \Longrightarrow \qquad \left|\frac{\langle\hat{A}\hat{B}\rangle}{\langle\hat{A}\rangle\langle\hat{B}\rangle} - 1\right| \ll 1 \tag{1.5}$$

which essentially states that the least the fluctuations, the better the approximation. This rule of thumb is necessary in order to interpret the physical relevance of numerical MFT results. We will come back to this in the end, for now assuming the validity of MFT approximation and deriving the mathematical results.

1.2.2 MFT treatment of the pure Hubbard model

The pure Hubbard model, $\hat{H}_t + \hat{H}_U$, offers a simple framework for MFT analysis of the AF phase: this is described in detail in App. **??**. The key passage is there given by the MFT approximation

$$\hat{n}_{i\uparrow}\hat{n}_{i\downarrow} \simeq \hat{n}_{i\uparrow}\langle\hat{n}_{i\downarrow}\rangle + \langle\hat{n}_{i\uparrow}\rangle\hat{n}_{i\downarrow} + (\text{constants}) \tag{1.6}$$

from which the AF structure is simply recovered. In terms of the discussion of the above paragraph, we have $\hat{A} \equiv \hat{n}_{i\uparrow}$ and $\hat{B} \equiv \hat{n}_{i\downarrow}$. However the "full" operator is given by

$$\hat{n}_{i\uparrow}\hat{n}_{i\downarrow} = \hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{j\downarrow}\hat{c}_{j\downarrow}\hat{c}_{i\uparrow}$$

whose terms we can contract in three different ways following Wick's Theorem:

$$\langle\hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\hat{c}_{i\uparrow}\rangle \simeq \underbrace{\langle\hat{c}^\dagger_{i\uparrow}\hat{c}^\dagger_{i\downarrow}\rangle\langle\hat{c}_{i\downarrow}\hat{c}_{i\uparrow}\rangle}_{\text{Cooper}} - \underbrace{\langle\hat{c}^\dagger_{i\uparrow}\hat{c}_{i\downarrow}\rangle\langle\hat{c}^\dagger_{i\downarrow}\hat{c}_{i\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle\hat{c}^\dagger_{i\uparrow}\hat{c}_{i\uparrow}\rangle\langle\hat{c}^\dagger_{i\downarrow}\hat{c}_{i\downarrow}\rangle}_{\text{Hartree}} \tag{1.7}$$

where the $-$ sign in the Fock term comes from fermionic anticommutation rules.

As a first approximation, the theorem is assumed to hold (which, in a BCS-like fashion, is equivalent to assuming for the ground-state to be a coherent state). The AF ground state reduces translational invariance and rotational symmetry, as explained in Tab. 1.1. Thus, of the three terms above, when considering the pure Hubbard model in AF phase:

- Cooper fluctuations are suppressed, because they break charge conservation;
- Similarly the Fock term is null as well because if $i = j$ and $\sigma' = \overline{\sigma}$ (as is for the local interaction, which contains the operator $\hat{n}_{i\uparrow}\hat{n}_{i\downarrow}$) the expectation values involved are describing a process breaking the survivor $\mathrm{U}^z(1)$ spin symmetry.

Thus, correctly, the Wick's decomposition of pure HM only involves Hartree-terms of Eq. (1.7), getting us back to (1.6). The distinction between the normal phase and the AF phase is given by whether the Hartree fluctuations break or not translational symmetry: AF is indeed a specific SDW ordering which weakens the latter. In general the three parts of the decomposition (Hartree, Fock, Cooper) need to be considered altogether: this is what is done in the next chapters. What said here extends naturally to the EHM: the relevant contributions for each phase given by the local repulsion $\hat{U}$ in the EHM are listed in Tab. 1.3.

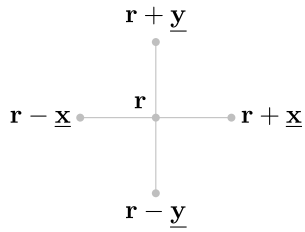


Figure 1.2 | Schematic representation of the four NNs of a given site i for a planar square lattice.

1.3 The non-local attraction $\hat{H}_V$ as a source of symmetry-breaking interactions

Let us now extend the discussion to the EHM and consider the NN non-local attraction,

$$\hat{H}_V \equiv -V \sum_{\langle \mathbf{r}\mathbf{r}' \rangle} \sum_{\sigma\sigma'} \hat{n}_{\mathbf{r}\sigma} \hat{n}_{\mathbf{r}'\sigma'} \tag{1.8}$$

where we changed notation, and replaced the site index i with its 2D position on the lattice

$$i \quad \rightarrow \quad \mathbf{r} = (x, y)$$

Let also the lattice versors,

$$\underline{\mathbf{x}} \equiv (1, 0) \qquad \underline{\mathbf{y}} \equiv (0, 1)$$

as in Fig. 1.2.

Evidently the hamiltonian can be decomposed in various spin terms,

$$\begin{aligned} \hat{H}_V &= \sum_{\sigma\sigma'} \hat{H}_V^{\sigma\sigma'} \\ &= \underbrace{\hat{H}_V^{\uparrow\uparrow} + \hat{H}_V^{\downarrow\downarrow}}_{\text{Same-spin}} + \underbrace{\hat{H}_V^{\uparrow\downarrow} + \hat{H}_V^{\downarrow\uparrow}}_{\text{Opposite-spin}} \end{aligned} \tag{1.9}$$

The role of said terms is crucial in establishing the pairing channel of the dominant physical phase. As an example, the s.s. terms can be seen as a possible source of triplet pairing superconductivity. Next section are devoted to analyze said terms and derive a simple analytical result in reciprocal space.

1.3.1 Real space description

Let us start from the real space form of Eq. (1.9). To carry out a summation over nearest neighbors $\langle \mathbf{r}\mathbf{r}' \rangle$ of a square lattice means precisely to sum over all links of the lattice. Then we can identify the generic opposite-spin (o.s.) term $\hat{H}_V^{\sigma\overline{\sigma}}$ as the one collecting the σ operators of sublattice $\mathcal{S}_a$ and $\overline{\sigma}$ operators of sublattice $\mathcal{S}_b$. The o.s. non-local interactions can be written as a sum of terms over just one of the two sublattices $\mathcal{S}_a$ and $\mathcal{S}_b$. Define the site hamiltonian

$$\hat{h}_V^{(\mathbf{r})} \equiv -V \sum_{\boldsymbol{\delta}} \left(\hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow} + \hat{n}_{\mathbf{r}\uparrow} \hat{n}_{\mathbf{r}-\boldsymbol{\delta}\downarrow} \right) \quad \text{with} \quad \boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\} \tag{1.10}$$

Then, it follows,

$$\begin{aligned} \hat{H}_V^{(\text{o.s.})} &= \overbrace{\sum_{\mathbf{r}\in\mathcal{S}_a} \hat{h}_V^{(\mathbf{r})}}^{\hat{H}_V^{\uparrow\downarrow}} + \overbrace{\sum_{\mathbf{r}\in\mathcal{S}_b} \hat{h}_V^{(\mathbf{r})}}^{\hat{H}_V^{\downarrow\uparrow}} \\ &= \sum_{\mathbf{r}\in\mathcal{S}} \hat{h}_V^{(\mathbf{r})} \end{aligned} \tag{1.11}$$

Sector	Term	Normal phase	Antiferromagnet	Superconductor
o.s.	Hartree			
	Fock	Prohibited	Prohibited	Prohibited
	Cooper	Prohibited	Prohibited	
s.s.	Hartree			
	Fock			
	Cooper	Prohibited	Prohibited	

Table 1.4 Summary of Wick's terms selection rules and relevant contributions to the effective hamiltonian for each phase limitedly to $\hat{H}_V$, assuming preservation of $\mathrm{U}^z(1)$ symmetry.

Here the notation of Fig. **??** is used. For each site $\mathbf{r}$ in a given sublattice, the nearest neighbors sites are four – all in the other sublattice. Similarly, the same-spin (s.s.) hamiltonian decomposes as

$$\hat{H}_V^{(\text{s.s.})} = -V \sum_{\mathbf{r}\in\mathcal{S}_a} \sum_{\boldsymbol{\delta},\sigma} \left(\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma} \right) \quad \text{with} \quad \boldsymbol{\delta} \in \{\underline{\mathbf{x}}, \underline{\mathbf{y}}\}$$

Opposite-spin terms. Consider first the o.s. terms of Eq. (1.10): take e.g. the NNs term $\hat{n}_{i\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}$. As in Eq. (1.7), taking Wick's contractions,

$$\begin{aligned}
\langle \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \\
&\simeq \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\downarrow}\rangle}_{\text{Hartree}}
\end{aligned}$$

Identical decompositions are given for all others NNs. Of the three terms above:

- The Cooper term breaks $\mathrm{U}^c(1)$ charge symmetry, allowing for supeconducting instabilities;
- The Fock term breaks $\mathrm{U}^z(1)$ symmetry, because it accounts for a site hop *plus* spin flip process;
- The Hartree term can reduce both translational invariance and spin rotational symmetry.

As a general rule of thumb, throughout this text we will discard the o.s. Fock term eveywhere, preserving a robust $\mathrm{U}^z(1)$ symmetry. For the normal state as well as for the AF instability only the Hartree term is to be accounted; for the SC phase also the Cooper term contributes. Note that for superconducting instabilities, due to superexchange mechanism (as explained in App. **??**) the o.s. term accounts for singlet pairing as well as zero-projection triplet pairing. Which channel is preferred, is a matter of thermodynamic advantage.

Same-spin terms. Consider then the same-spin terms: take e.g. the term $\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}$. As above,

$$\begin{aligned}
\langle \hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle &= \langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle \\
&= \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle\langle \hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Cooper}} - \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle}_{\text{Fock}} + \underbrace{\langle \hat{c}^\dagger_{\mathbf{r}\uparrow}\hat{c}_{\mathbf{r}\uparrow}\rangle\langle \hat{c}^\dagger_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\hat{c}_{\mathbf{r}+\boldsymbol{\delta}\uparrow}\rangle}_{\text{Hartree}}
\end{aligned}$$

Identical consideration as in the above paragraph hold for each term. The only difference with the o.s. terms is given by the Fock term: since the spin-flip process is absent, now the Fock fluctuations actually contribute as an effective NN hopping term. In other words, this term does not break $\mathrm{U}^z(1)$ symmetry and thus is perfectly legitimate in each phase. Tab. 1.4 summarizes the selection rules hereby presented.

1.3.2 Fourier transform of the model

It is useful to derive analytically the reciprocal-space form of the three hamiltonian parts. By the sign $\overset{\mathcal{F}}{=}$, we indicate a passage where a Fourier-transform is performed. $\mathcal{F}$-transform is obtained according to the convention Fourier transform it according to the convention

$$\hat{c}_{\mathbf{r}\sigma} \overset{\mathcal{F}}{=} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in\mathrm{BZ}} e^{-i\mathbf{k}\cdot\mathbf{r}} \hat{c}_{\mathbf{k}\sigma} \tag{1.12}$$

Kinetic part. The kinetic part is the easiest to transform. Apply the definition,

$$\begin{aligned} -t \sum_{\langle\mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \left[\hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}_{\mathbf{r}'\sigma} + \text{h.c.}\right] &\overset{\mathcal{F}}{=} -t \sum_{\langle\mathbf{r}\mathbf{r}'\rangle} \sum_{\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}\in\mathrm{BZ}} e^{i\mathbf{k}\cdot\mathbf{r}} \hat{c}^\dagger_{\mathbf{k}\sigma} \frac{1}{\sqrt{L_x L_y}} \sum_{\mathbf{k}'\in\mathrm{BZ}} e^{-i\mathbf{k}'\cdot\mathbf{r}'} \hat{c}_{\mathbf{k}'\sigma} + \text{h.c.} \\ &= -2t \sum_{\mathbf{k},\mathbf{k}'} \sum_{\sigma} \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}'\sigma} \frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{S}/2} e^{i(\mathbf{k}-\mathbf{k}')\cdot\mathbf{r}} \left[e^{ik'_x} + e^{-ik'_x} + e^{ik'_y} + e^{-ik'_y}\right] \\ &= \sum_{\mathbf{k}\in\mathrm{BZ}} \sum_{\sigma} \epsilon_{\mathbf{k}} \hat{c}^\dagger_{\mathbf{k}\sigma}\hat{c}_{\mathbf{k}\sigma} \end{aligned} \tag{1.13}$$

where the bare bands were defined,

$$\epsilon_{\mathbf{k}} \equiv -2t\left(\cos k_x + \cos k_y\right) \tag{1.14}$$

Non-local attraction. Consider a generic bond, say, the one connecting NNs $\mathbf{r}$ and $\mathbf{r}\pm\boldsymbol{\delta}$. Then:

$$\begin{aligned} \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'} &= \hat{c}^\dagger_{\mathbf{r}\sigma}\hat{c}^\dagger_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\pm\boldsymbol{\delta}\sigma'}\hat{c}_{\mathbf{r}\sigma} \\ &\overset{\mathcal{F}}{=} \frac{1}{(L_x L_y)^2} \sum_{\mathbf{k}_1,\cdots,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} e^{\pm i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}} \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma} \end{aligned}$$

Then, the interaction at site $\mathbf{r}$, spin σ with its NNs at spin σ' – indicated as $(\mathbf{r}\sigma\sigma')$ – is given by

$$\begin{aligned} (\mathbf{r}\sigma\sigma') &= -V \sum_{\boldsymbol{\delta}\in\{\underline{\mathbf{x}},\underline{\mathbf{y}}\}} \left[\hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}+\boldsymbol{\delta}\sigma'} + \hat{n}_{\mathbf{r}\sigma}\hat{n}_{\mathbf{r}-\boldsymbol{\delta}\sigma'}\right] \\ &\overset{\mathcal{F}}{=} -\frac{V}{(L_x L_y)^2} \sum_{\boldsymbol{\delta}} \sum_{\mathbf{k}_1,\cdots,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \\ &\qquad\qquad \times \left(e^{i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}} + e^{-i(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}}\right) \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma} \\ &= -\frac{2V}{(L_x L_y)^2} \sum_{\boldsymbol{\delta}} \sum_{\mathbf{k}_1,\cdots,\mathbf{k}_4} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} \cos\left[(\mathbf{k}_2-\mathbf{k}_3)\cdot\boldsymbol{\delta}\right] \hat{c}^\dagger_{\mathbf{k}_1\sigma}\hat{c}^\dagger_{\mathbf{k}_2\sigma'}\hat{c}_{\mathbf{k}_3\sigma'}\hat{c}_{\mathbf{k}_4\sigma} \end{aligned}$$

The full non-local interaction is given by summing over all sites of $\mathcal{S}/2$ (which is, one sublattice of $\mathcal{S}$). This gives back momentum conservation,

$$\frac{1}{L_x L_y} \sum_{\mathbf{r}\in\mathcal{S}/2} e^{i[(\mathbf{k}_1+\mathbf{k}_2)-(\mathbf{k}_3+\mathbf{k}_4)]\cdot\mathbf{r}} = \frac{1}{2}\delta_{\mathbf{k}_1+\mathbf{k}_2=\mathbf{k}_3+\mathbf{k}_4}$$

Let $\mathbf{k}_1+\mathbf{k}_2 = \mathbf{k}_3+\mathbf{k}_4 = \mathbf{K}$, and define $\mathbf{k}$, $\mathbf{k}'$ such that

$$\mathbf{k}_1 \equiv \mathbf{K}+\mathbf{k} \qquad \mathbf{k}_2 \equiv \mathbf{K}-\mathbf{k} \qquad \mathbf{k}_3 \equiv \mathbf{K}-\mathbf{k}' \qquad \mathbf{k}_4 \equiv \mathbf{K}+\mathbf{k}' \qquad \delta\mathbf{k} \equiv \mathbf{k}-\mathbf{k}'$$

Sums over these variables must be intended as over the Brillouin Zone (BZ). Hence:

$$\begin{aligned} \hat{H}_V &= \sum_{\mathbf{r}\in\mathcal{S}_a} \sum_{\sigma\sigma'} (\mathbf{r}\sigma\sigma') \\ &\overset{\mathcal{F}}{=} -\frac{V}{L_x L_y} \sum_{\sigma\sigma'} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \cos\left(\delta\mathbf{k}\cdot\boldsymbol{\delta}\right) \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \\ &= -\frac{V}{L_x L_y} \sum_{\sigma\sigma'} \sum_{\mathbf{K},\mathbf{k},\mathbf{k}'} \left[\cos\left(\delta k_x\right) + \cos\left(\delta k_y\right)\right] \hat{c}^\dagger_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^\dagger_{\mathbf{K}-\mathbf{k}\sigma'}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma'}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} \end{aligned} \tag{1.15}$$

which decomposes in s.s. and o.s. channels as:

$$\hat{H}_V \overset{\mathcal{F}}{=} \overbrace{-\frac{V}{L_x L_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}^{\text{s.s.}}$$
$$\underbrace{-\frac{V}{L_x L_y}\sum_{\sigma}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}}_{\text{o.s.}}$$

The o.s. part can be simplified: since

$$\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma} = \hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\overline{\sigma}}\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}\hat{c}_{\mathbf{K}+\mathbf{k}'\sigma}\hat{c}_{\mathbf{K}-\mathbf{k}'\overline{\sigma}}$$

due to fermionic anti-commutation rules, and since the composite transform

$$\mathbf{k}\to-\mathbf{k}\qquad \mathbf{k}'\to-\mathbf{k}'$$

leaves the form factor $\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)$ invariant, we reduce the spin sum to two times the $\sigma=\uparrow$ part,

$$\hat{H}_V^{(\text{o.s.})} = -\frac{2V}{L_x L_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{1.16}$$

whiel the explicit s.s. sector is given immediately by:

$$\hat{H}_V^{(\text{s.s.})} = -\frac{V}{L_x L_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\left[\cos\left(\delta k_x\right)+\cos\left(\delta k_y\right)\right]$$
$$\times\left[\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\uparrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\uparrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow}+\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\downarrow}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\downarrow}\right] \tag{1.17}$$

The result here obtained will be used various times in next chapters. Note that different Wick contraction schemes will be referred to as:

$$c^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}c^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma'}c_{\mathbf{K}-\mathbf{k}'\sigma'}c_{\mathbf{K}+\mathbf{k}'\sigma}\qquad \text{Cooper contraction} \tag{1.18}$$
$$c^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}c^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma'}c_{\mathbf{K}-\mathbf{k}'\sigma'}c_{\mathbf{K}+\mathbf{k}'\sigma}\qquad \text{Fock contraction} \tag{1.19}$$
$$c^{\dagger}_{\mathbf{K}+\mathbf{k}\sigma}c^{\dagger}_{\mathbf{K}-\mathbf{k}\sigma'}c_{\mathbf{K}-\mathbf{k}'\sigma'}c_{\mathbf{K}+\mathbf{k}'\sigma}\qquad \text{Hartree contraction} \tag{1.20}$$

When contracting in the Fock channel, a minus sign must be included.

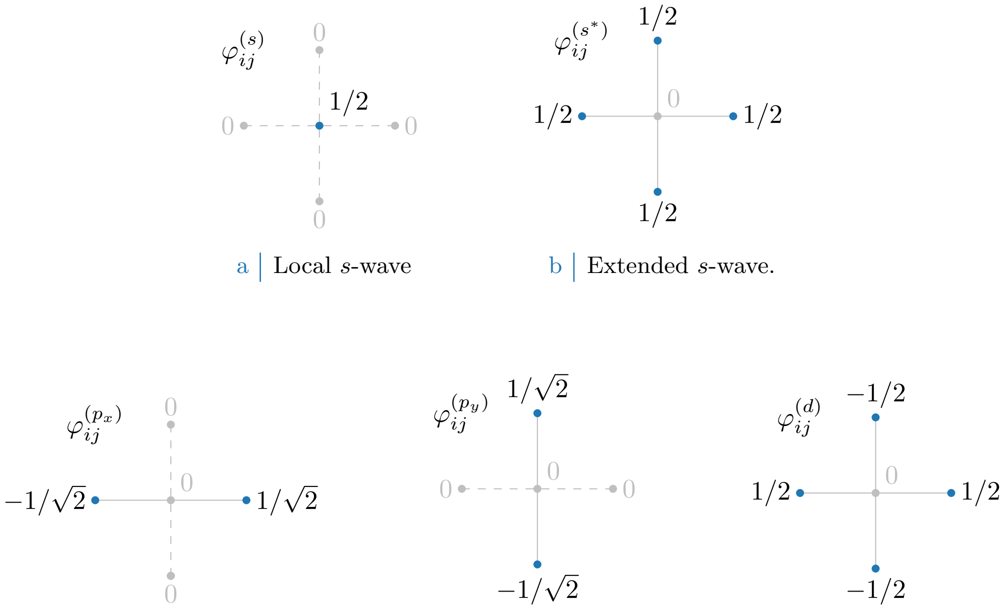
Local repulsion. A somewhat identical argument can be carried out for the local interaction,

$$\hat{H}_U = U\sum_{\mathbf{r}\in\mathcal{S}}\hat{n}_{\mathbf{r}\uparrow}\hat{n}_{\mathbf{r}\downarrow}$$

The only difference here is made by the fact that the interaction is repulsive on-site, thus the final structure factor in the Fourier transform disappears as well as the minus sign,

$$\hat{H}_U \overset{\mathcal{F}}{=} \frac{U}{L_x L_y}\sum_{\mathbf{K},\mathbf{k},\mathbf{k}'}\hat{c}^{\dagger}_{\mathbf{K}+\mathbf{k}\uparrow}\hat{c}^{\dagger}_{\mathbf{K}-\mathbf{k}\downarrow}\hat{c}_{\mathbf{K}-\mathbf{k}'\downarrow}\hat{c}_{\mathbf{K}+\mathbf{k}'\uparrow} \tag{1.21}$$

When contracting these fermionic operators, identical considerations as for the non-local counterpart (1.15) hold. Note the similarity of this Fourier transform with the o.s. sector of the non-local attraction, given in Eq. (1.15). With this, we conclude the general introduction to the model and pass to the setup of the Hartree-Fock (HF) algorithm.



a | Local s-wave b | Extended s-wave.

c | p_x-wave. d | p_y-wave. e | $d_{x^2-y^2}$-wave.

Figure 1.3 | Form factors at different topologies, as listed in Tab. 1.5. In figures five sites are represented: the hub site i and its four NN. Solid lines represent non-zero values for $\varphi_{\boldsymbol{\delta}}$, while dashed lines represent vanishing factors.

1.4 Square lattice spatial symmetry channels

In next sections we introduce spatial symmetry structures and discuss their relation with point groups. We will aways refer to various spatial structures as "symmetry channels", throughout the system is able to establish a given order parameter defined in space.

Looking at $\hat{H}$, the main difference between its two-body interactions $\hat{H}_U$ and $\hat{H}_V$ of course relies in locality (first) and non-locality (second). Both operators in principle are able to break C_4 symmetry, as will later be discussed extensively.

For the 2D square lattice, various spatial symmetries are possible. Let i be a specific site, and consider a generic complex function g_{ij} whose domain is given by the site itself and its four NNs (the domain concides with the points of Fig. 1.2). The function can be written in the C_4 basis

$$g_{ij} = \sum_{\gamma} g^{(\gamma)} \varphi_{ij}^{(\gamma)}$$

where $g^{(\gamma)} \in \mathbb{C}$ are the g_{ij} symmetry channels coefficients, while $\varphi_{ij}^{(\gamma)}$ are the form factors listed in Tab. 1.5 and plotted in Fig. 1.4, a simple orthonormal rearrangement of the harmonics basis.

When dealing with phases which respect translational symmetry, or at most reduce it to a weaker symmetry by shrinking the Brillouin Zone, it's useful to work in reciprocal space, as discussed in Sec. 1.3. As said, a particular phase can show specific spatial symmetries, and so will the Hartree-Fock parameters. The key point to notice is that self-consistency equations involve sums (or integrals) to be computed over fermionic states, which can be heavily simplified by employing said symmetries. Consider the form factors of Tab. 1.5, and their Fourier transforms of Tab. 1.6. They satisfy the relations:

$$\frac{1}{L_x L_y} \sum_{ij} \varphi_{ij}^{(\gamma)} \varphi_{ij}^{(\gamma)*} = \delta_{\gamma\gamma'} \qquad \frac{1}{L_x L_y} \sum_{\mathbf{k}} \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}}^{(\gamma)*} = \delta_{\gamma\gamma'}$$

If these symmetries are developed by the HFPs, it's of no use to integrate over the entire BZ, since different regions linked by periodicity of the form factors will lead to redundant calculations.

Structure	Form factor	Graph
s-wave	$\varphi^{(s)}_{\mathbf{r}\mathbf{r}'} = \delta_{\mathbf{r}\mathbf{r}'}$	Fig. 1.3a
Extended s-wave	$\varphi^{(s^*)}_{\mathbf{r}\mathbf{r}'} = \frac{1}{2}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 1.3b
p_x-wave	$\varphi^{(p_x)}_{\mathbf{r}\mathbf{r}'} = \frac{1}{\sqrt{2}}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}}\right]$	Fig. 1.3c
p_y-wave	$\varphi^{(p_y)}_{\mathbf{r}\mathbf{r}'} = \frac{1}{\sqrt{2}}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 1.3d
$d_{x^2-y^2}$-wave	$\varphi^{(d)}_{\mathbf{r}\mathbf{r}'} = \frac{1}{2}\left[\delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{x}}} + \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{x}}} - \delta_{\mathbf{r}'=\mathbf{r}+\underline{\mathbf{y}}} - \delta_{\mathbf{r}'=\mathbf{r}-\underline{\mathbf{y}}}\right]$	Fig. 1.3e

Table 1.5 | Form factors for a function defined on a square lattice, with a prefactor included for normalization. We use lattice vector notation here. The last column indicates the graph representation of each contribution given in Fig. 1.3. Subscript x^2-y^2 is omitted for notational clarity.

Structure	Structure factor	Graph
s-wave	$\varphi^{(s)}_{\mathbf{k}} = 1$	Fig. 1.3a
Extended s-wave	$\varphi^{(s^*)}_{\mathbf{k}} = \cos k_x + \cos k_y$	Fig. 1.3b
p_x-wave	$\varphi^{(p_x)}_{\mathbf{k}} = i\sqrt{2}\sin k_x$	Fig. 1.3c
p_y-wave	$\varphi^{(p_y)}_{\mathbf{k}} = i\sqrt{2}\sin k_y$	Fig. 1.3d
$d_{x^2-y^2}$-wave	$\varphi^{(d)}_{\mathbf{k}} = \cos k_x - \cos k_y$	Fig. 1.3e

Table 1.6 | Structure factors derived from the correlation structures of Tab. 1.5. The functions hereby defined are orthonormal, and are plotted in Fig- 1.4.

Finally, a very useful detail is given by the following argument. Consider the equation

$$\cos(\delta k_x) + \cos(\delta k_y) = \cos k_x \cos k'_x + \sin k_x \sin k'_x + \cos k_y \cos k'_y + \sin k_y \sin k'_y$$

For the sake of readability, the notations

$$c_\ell \equiv \cos k_\ell \qquad s_\ell \equiv \sin k_\ell \qquad c'_\ell \equiv \cos k'_\ell \qquad s'_\ell \equiv \sin k'_\ell$$

are used. Group the four terms above,

$$\underbrace{\left(c_x c'_x + c_y c'_y\right)}_{\text{Symmetric}} + \underbrace{\left(s_x s'_x + s_y s'_y\right)}_{\text{Anti-symmetric}} \tag{1.22}$$

The first two exhibit inversion symmetry for both arguments $\mathbf{k}$, $\mathbf{k}'$; the second two exhibit anti-symmetry. Decoupling the symmetric part, we obtain

$$c_x c'_x + c_y c'_y = \frac{1}{2}(c_x + c_y)(c'_x + c'_y) + \frac{1}{2}(c_x - c_y)(c'_x - c'_y)$$

which finally gives:

$$\begin{aligned}
\cos(\delta k_x) + \cos(\delta k_y) = {} & \frac{1}{2}(c_x + c_y)(c'_x + c'_y) && (s^*\text{-wave}) \\
& + s_x s'_x && (p_x\text{-wave}) \\
& + s_y s'_y && (p_y\text{-wave}) \\
& + \frac{1}{2}(c_x - c_y)(c'_x - c'_y) && (d_{x^2-y^2}\text{-wave})
\end{aligned}$$

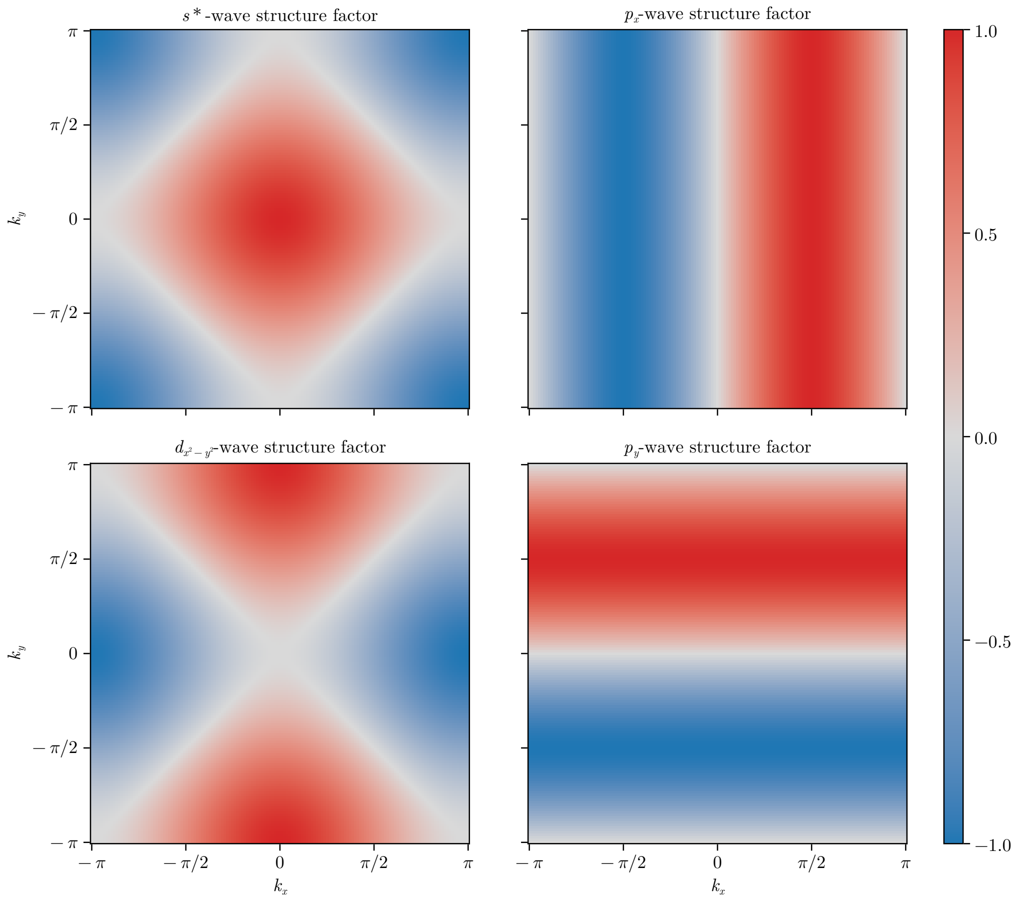


Figure 1.4 Representation of the various structure factors listed in Tab. 1.6 on the BZ. For the s^* and $d_{x^2-y^2}$-wave factors, the real part is plotted; for the p_ℓ factors, the imaginary part is plotted.

or, in a more compact form

$$\cos(\delta k_x) + \cos(\delta k_y) = \frac{1}{2}\sum_\gamma \varphi_{\mathbf{k}}^{(\gamma)} \varphi_{\mathbf{k}'}^{(\gamma)*} \quad \text{for } \gamma \in \{s^*, p_x, p_y, d_{x^2-y^2}\} \tag{1.23}$$

$$= \frac{1}{2}\sum_\gamma \varphi_{\mathbf{k}}^{(\gamma)*} \varphi_{\mathbf{k}'}^{(\gamma)} \tag{1.24}$$

This decomposition will become particularly handy in later calculations.

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